Core Learning Objective of Phase 3

Understand and implement the **ETL workflow (Extract → Transform → Load)** using PySpark.

□ What Students Should Learn (Deep Level)

□ 1. Data is NOT clean by default

This is the first big mindset shift.

They should realize:

- real data has nulls
- incorrect values exist
- schema may not be perfect

□ Learning:

- `dropna()`
- `fillna()`
- filtering invalid records

□ 2. Ingestion (Extract step)

Students should become comfortable with:

- reading data from files
- understanding schema

Example:

```
spark.read.option("header","true").csv(...)
```

□ Learning:

- how data enters the system

□ 3. Schema Awareness

They should start asking:

- What are column types?
- Are they correct?
- Do I need casting?

□ This is the beginning of **data modeling awareness**

□ 4. Transformation Thinking

Instead of writing isolated queries, they should think:

```
read → clean → filter → join → aggregate
```

□ Learning:

- chaining transformations
- step-by-step data flow

□ 5. Pipeline Thinking (MOST IMPORTANT)

This is the heart of Phase 3.

They should understand:

A pipeline is a sequence of transformations applied to data

Not:

- random queries
  But:
- structured flow

□ 6. Order of Operations Matters

They should learn:

- clean BEFORE join
- filter BEFORE aggregation
- validate BEFORE processing

 This is real engineering thinking.

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7. SQL → PySpark Mapping (Reinforced)

By now they should be able to confidently map:

SQL	PySpark
SELECT	select()
WHERE	filter()
GROUP BY	groupBy()
JOIN	join()

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8. Basic Data Validation Mindset

They should start thinking:

- Are keys null?
- Are values valid?
- Are duplicates present?

 Even simple checks build strong habits.

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9. Confidence Working with Real Data

Compared to Phase 1 & 2:

- Phase 1 → toy data
- Phase 2 → sample files
- Phase 3 → real-world workflow

 They should feel:
"I can take raw data and process it"

What You Should Be Able to Do After Phase 3

By the end, a student should be able to:

- read a dataset
- inspect schema
- clean data
- filter invalid records
- join datasets
- aggregate results
- think in pipeline steps

What You Should NOT Focus On

- not perfection
- not advanced optimization
- not memorizing syntax

 Focus is:
process understanding

One-Line Summary

Phase 3 is about learning how to take raw data, clean it, and transform it step-by-step into meaningful output using a structured pipeline.